Lesson 7.4 - Probability of an Exponential Distribution

> **Video:** Probability Exponential Distribution Problems by The Organic Chemistry Tutor **Chapter:** 7 - Continuous Distributions

Formulas

Quantity	Formula
Rate parameter	$\lambda = \dfrac{1}{\mu}$
PDF	$f(x) = \lambda e^{-\lambda x},\ x \geq 0$
Area to the **left**	$P(X < x) = 1 - e^{-\lambda x}$
Area to the **right**	$P(X > x) = e^{-\lambda x}$
Area **between** a and b	$P(a < X < b) = P(X < b) - P(X < a) = e^{-\lambda a} - e^{-\lambda b}$

($e \approx 2.71828$. The video once says "rate parameter = 1/λ"; it means $\lambda = 1/\mu$.)

Example - laptop lifespan

Laptops from company XYZ last **5 years** on average, and the lifespan follows an exponential distribution.

(a) Rate parameter

$$\lambda = \frac{1}{\mu} = \frac{1}{5} = 0.20 \text{ per year}$$

(b) PDF and graph

$$f(x) = \lambda e^{-\lambda x} = 0.20\, e^{-0.20x}$$

The graph starts at the **y-intercept** $f(0) = 0.20 = \lambda$ (since $e^0 = 1$) and decays toward 0 as x grows.

(c) Lasts less than 3 years (left area)

$$P(X < 3) = 1 - e^{-\lambda x} = 1 - e^{-0.20(3)} = 1 - e^{-0.6} \approx 0.4512 = 45.12\%$$

(d) Lasts more than 10 years (right area)

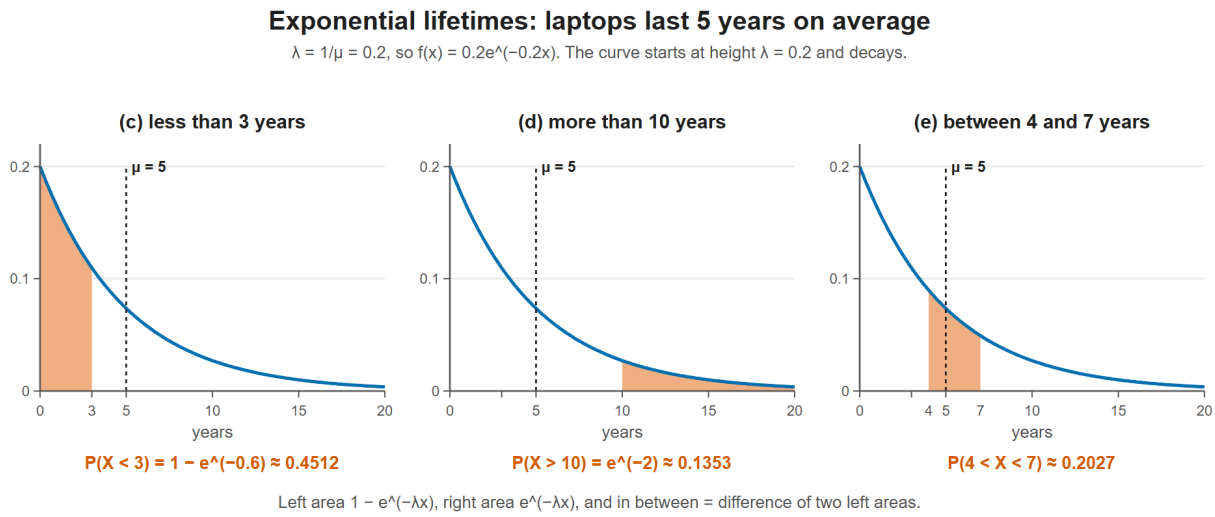
$$P(X > 10) = e^{-\lambda x} = e^{-0.20(10)} = e^{-2} \approx 0.1353 = 13.53\%$$

(e) Lasts between 4 and 7 years

The area left of 7 minus the area left of 4 leaves the region between them:

$$P(4 < X < 7) = P(X < 7) - P(X < 4) = \left(1 - e^{-1.4}\right) - \left(1 - e^{-0.8}\right) \approx 0.75340 - 0.55067 = 0.20273 \approx 20.3\%$$

(Equivalently, $e^{-0.8} - e^{-1.4} \approx 0.4493 - 0.2466 = 0.2027$.)



The three laptop questions as areas under f(x) = 0.2e^(−0.2x). The curve starts at λ = 0.2, and the mean lifetime is 5 years.

Key takeaways

- Find $\lambda = 1/\mu$ first. The PDF's y-intercept equals λ.
- Left area $1 - e^{-\lambda x}$; right area $e^{-\lambda x}$; between = difference of the left areas.
- The exponential distribution models **waiting times and lifetimes**. Short times are most likely, and long times become less and less likely.